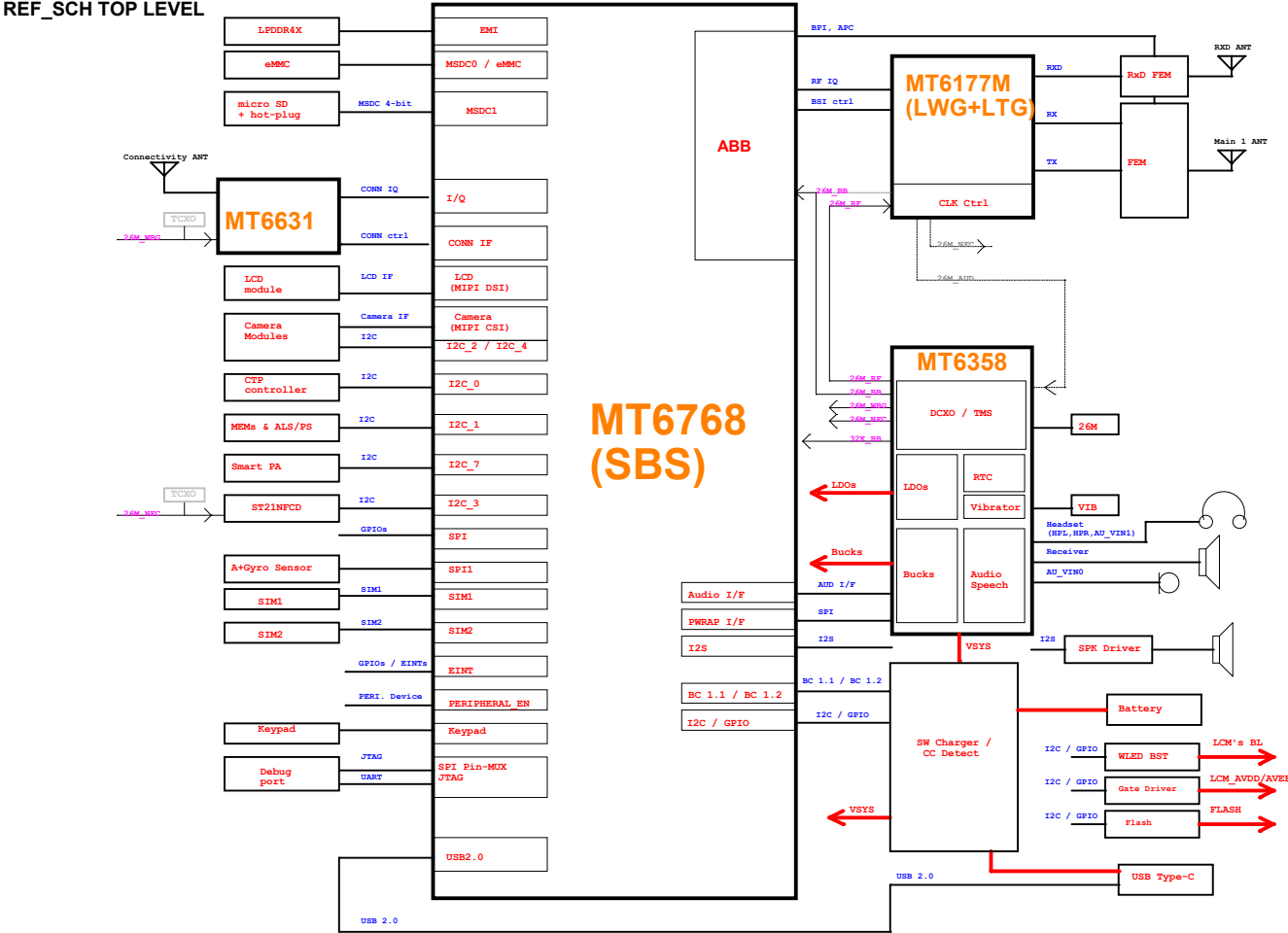
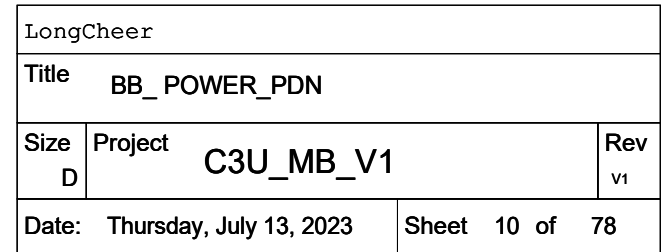
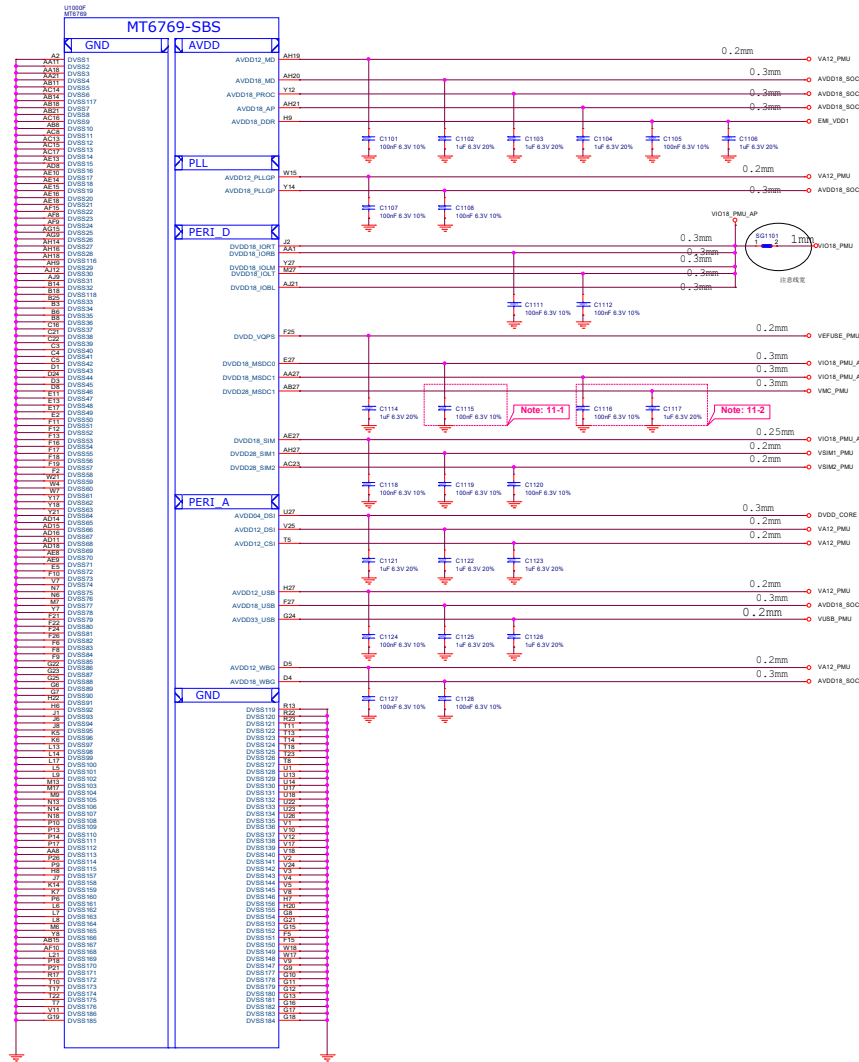




LongCheer		
Title 01_Block_Diagram		
Size D	Project C3U_MB_V1	Rev V1
Date: Thursday, July 13, 2023		Sheet 1 of 24

Note 10-1:	Differential pair of DVDD_GPU remote sense must be close to BB's ball. Remote sense trace with GND shielding to PMIC (Differential)
Note 10-2:	Differential pair of DVDD_PROC remote sense must be close to BB's ball. Remote sense trace with GND shielding to PMIC (Differential)
Note 10-3:	Differential pair of DVDD_MODEM remote sense must be close to BB's ball. Remote sense trace with GND shielding to PMIC (Differential)
Note 10-4:	Differential pair of DVDD_CORE remote sense must be close to BB's ball. Remote sense trace with GND shielding to PMIC (Differential)





Schematic design notice of "11_BB_POWER_IO" page:

Note 11-1: C1115 closed DVDD18_MSDC0 150mil

Note 11-2: C1116 closed DVDD18_MSDC1 150mil
C1117 closed DVDD28_MSDC1 150mil

AVDD12 :单个点走8mil,两路分支各16mil,总路径20mil

VIO18_PMU_AP:总路径40mil

AVDD18_SOC:总路径20mil

LongCheer

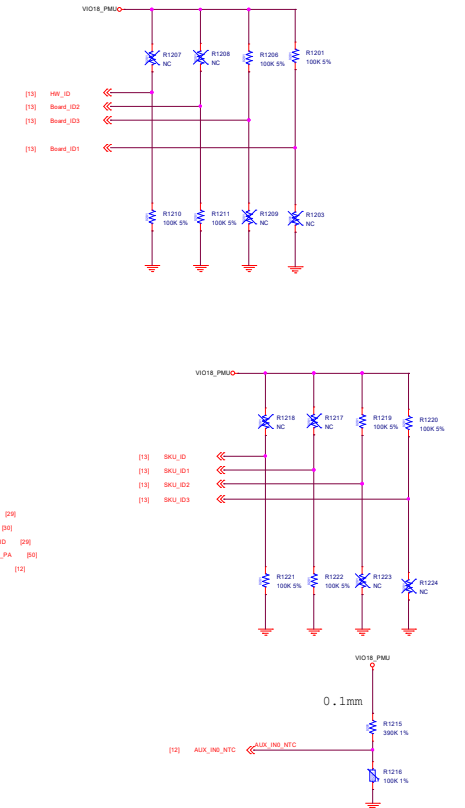
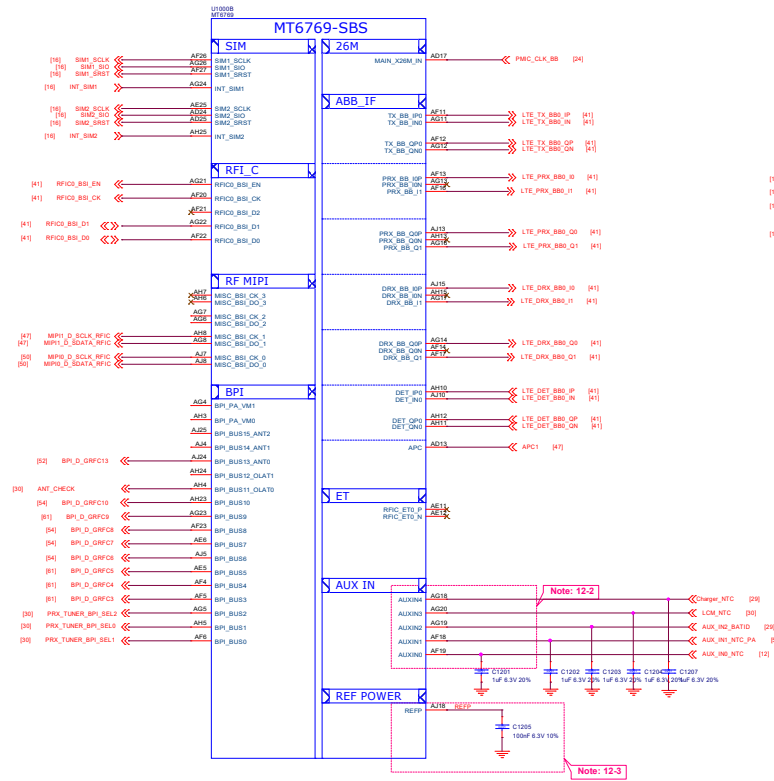
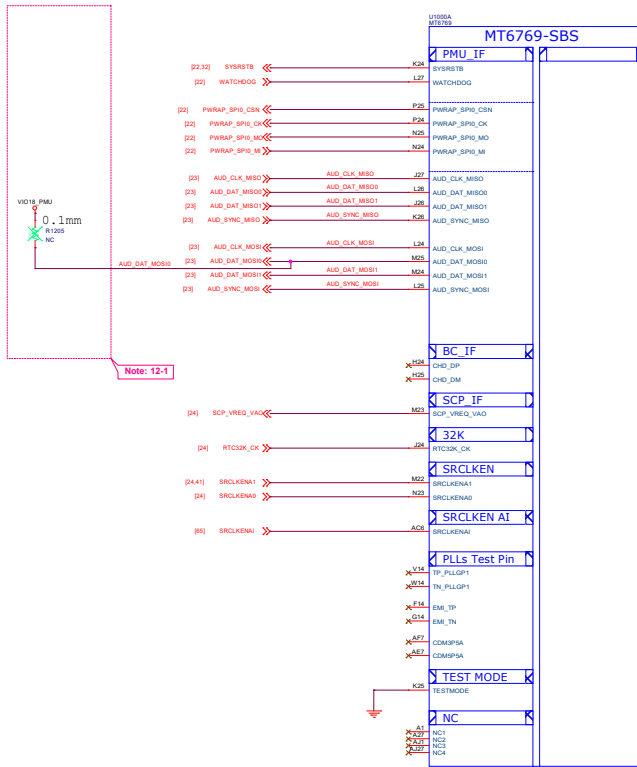
Title BB_POWER_IO

Size D Project C3U_MB_V1

Rev v1

Date: Thursday, July 13, 2023

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Schematic design notice of "12_BB_1" page:

Note 12-1: PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pins to select which interface will be the JTAG pin out

PWRAP_SPI0_CSN	AUD_DAT_MOSI0	JTAG Function
default=PU	default=PD	AP_JTAG MD_JTAG
HI	LO	N/A
HI	HI (by ext. PD)	SPI0 + EINT8 SPI2 + SPI3
LO	LO	SPI0 + EINT8
LO	HI (by ext. PD)	N/A

Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-3: The de-coupling cap. for REFP (AJ18 ball) have to be placed as close to BB as possible.

Note 12-4: HW pin for DDR type feature in bootstrap (refer to BB HW design Notice)

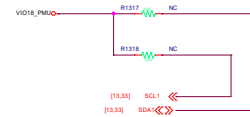
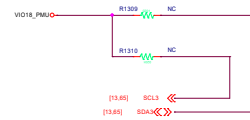
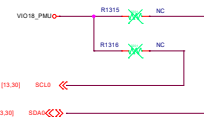
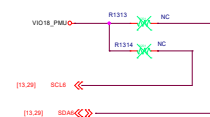
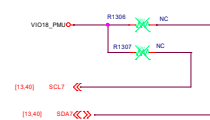
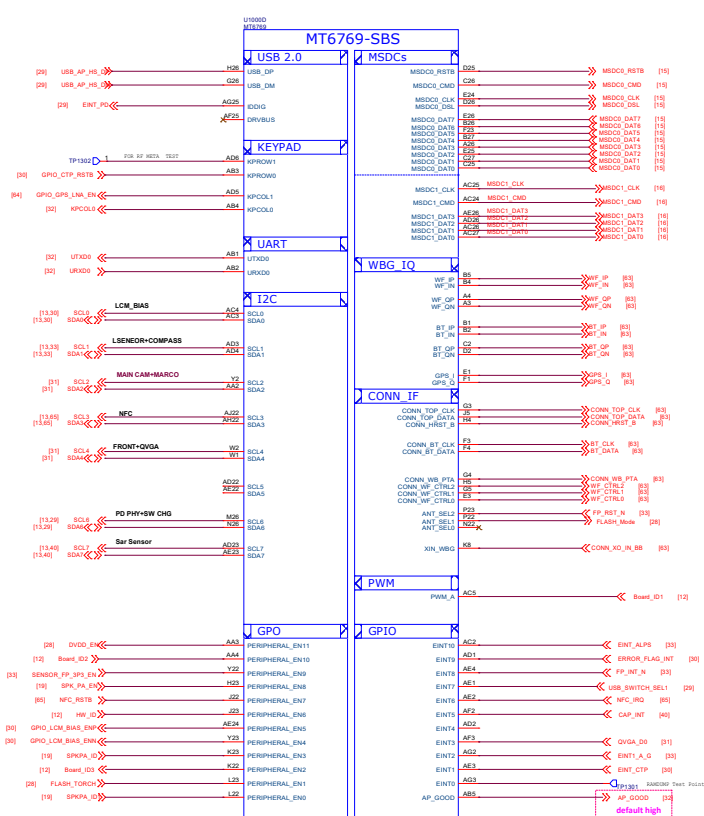
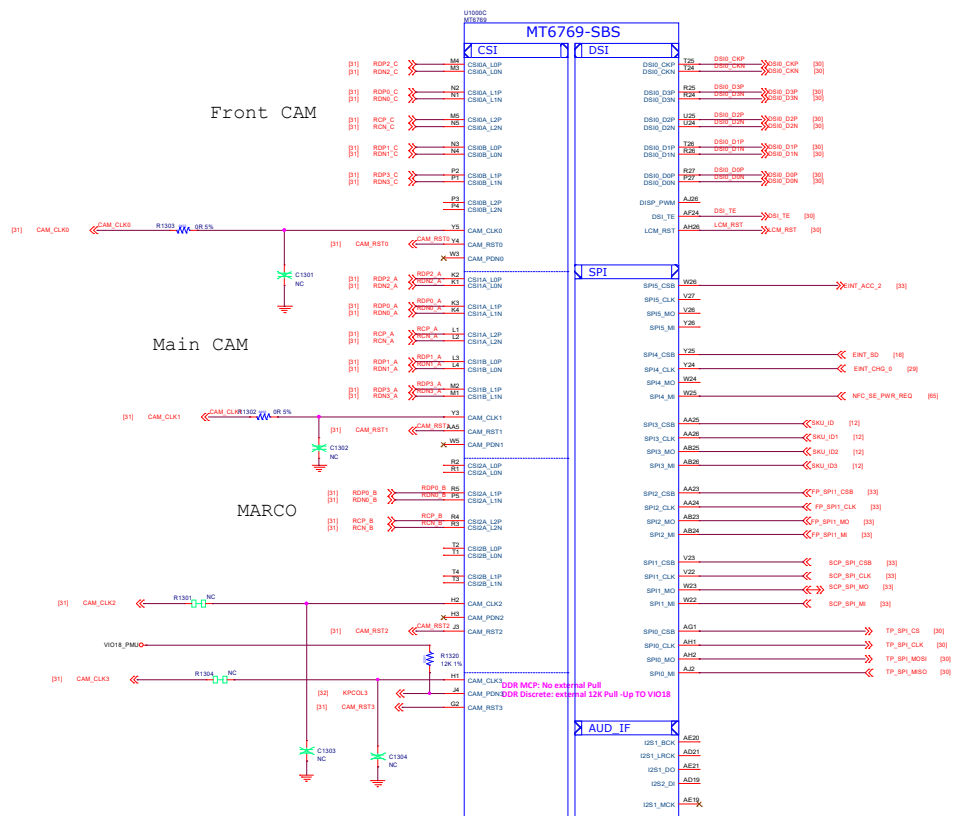
AUD_SYNC MISO	AUD_CLK MISO	CAM PDN3	PMIC 6358 voltage	DDR Type
default=PD	default=PD	default=PD	VDRAM1 / VDRAM2	DDR
No ext. pull	No ext. pull	No ext. pull	1.125v / 0.6v	LP4X eMCP
No ext. pull	12K pull to VIO18	No ext. pull	OFF / 1.8v	Reserved
12K pull to VIO18	No ext. pull	No ext. pull	1.225v / OFF	LP3 eMCP
12K pull to VIO18	12K pull to VIO18	No ext. pull	1.125v / 1.8v	Reserved
No ext. pull	No ext. pull	12K pull to VIO18	1.125v / 0.6v	LP4X discrete

Note 12-5: Charger must have D+/D- pin for charger type USB detection.
Charger must have at least 500mA USB current for All charger type.

LongCheer		
Title 12_BB_1		
Size D	Project C3U_MB_V1	Rev V1
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Thermistor to sense AP temperature

- R1216 must keep a distance about 6-8 mm away from AP and far from other heat sources 10 mm at least.
- The distance is the shortest distance from package edge to edge.



LongCheer

Title 13_BB_2

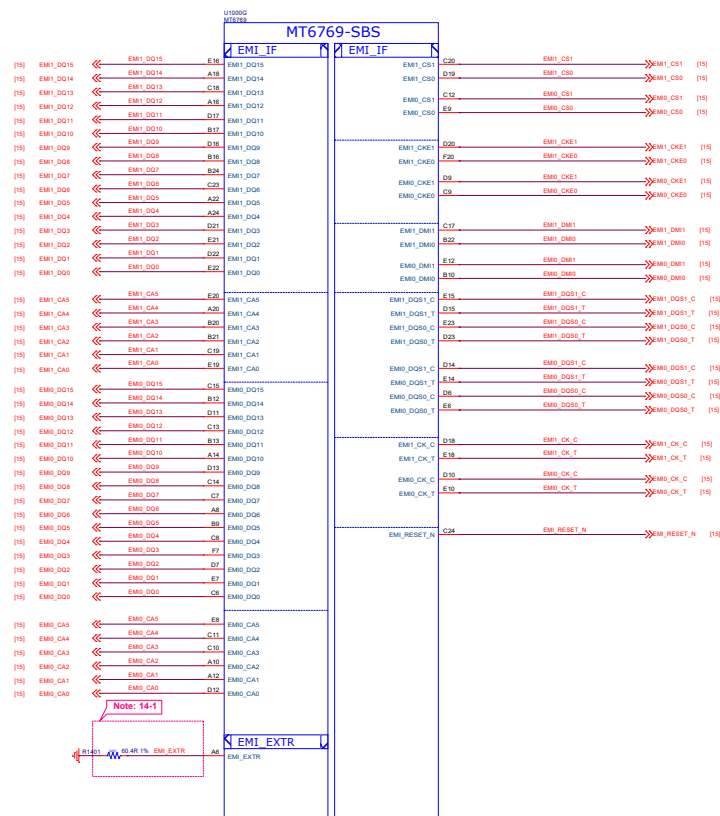
Size D Project

C3U_MB_V1

Rev v1

Date: Thursday, July 13, 2023

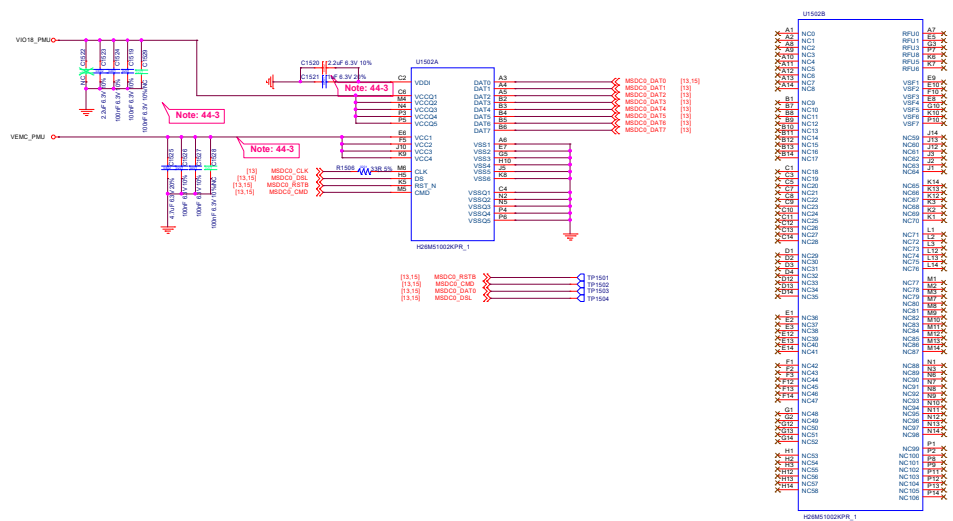
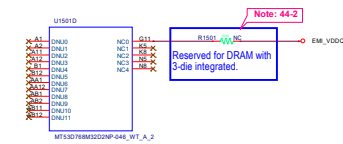
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Schematic design notice of "14_BB_3" page.

Note 14-1: R1401 please select 60.4 ohm (1%) resistor

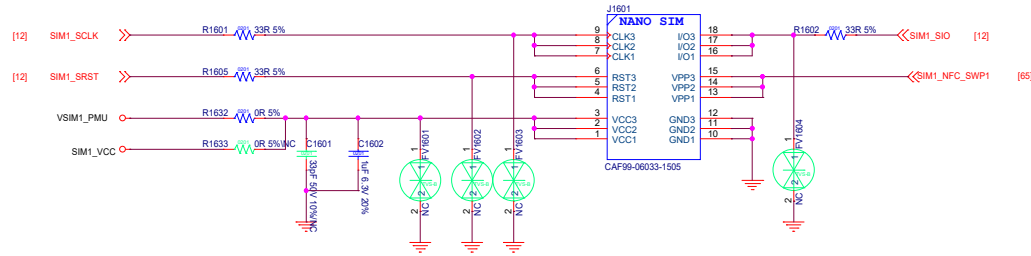
LongCheer		
Title 14_BB_3		
Size D	Project C3U_MB_V1	Rev V1
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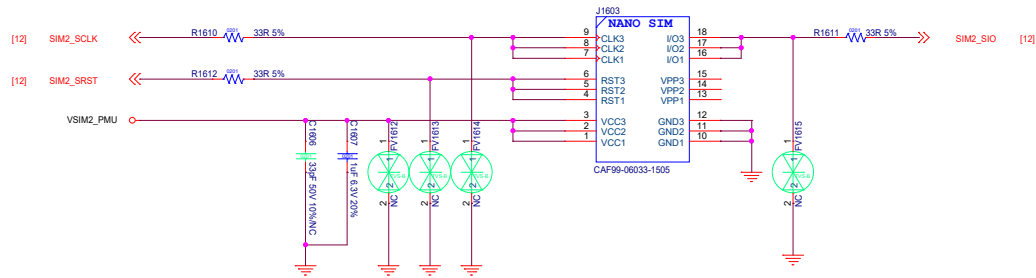
Note 44-4: VDD2 VDDQ decoupling cap: closed to DRAM ball.
For other cap for PMIC (>10uF, at PMIC page):
please also refer to MMD and layout guide for placement.

LongCheer			
Title Memory_eMMC_LPDDR4X			
Size D	Project C3U_MB_V1		Rev V1
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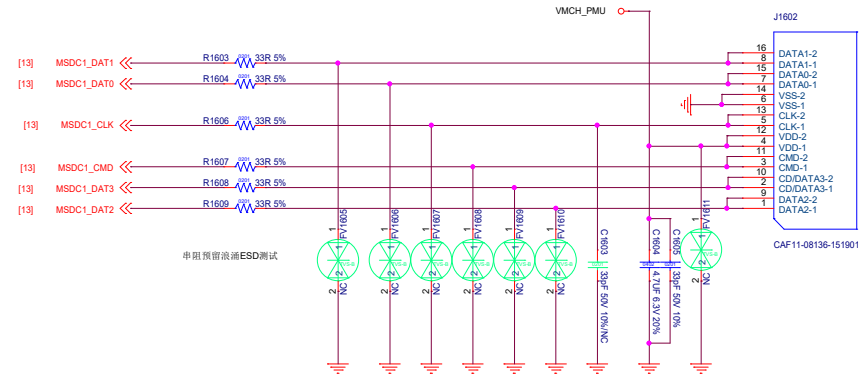
SIM1 PCB靠板边



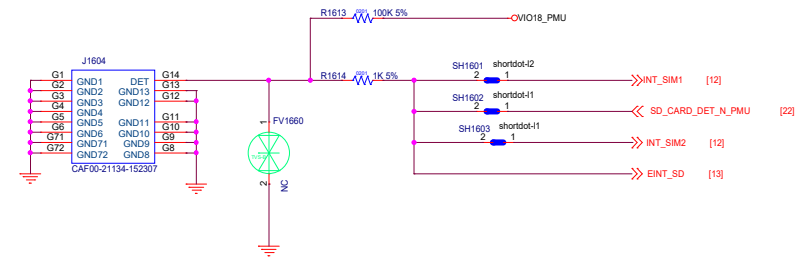
SIM2 PCB靠板内



SD CARD



Detect



Note 46-1:

Schematic design notice of "46_MEMORY_SD Card" page.

For better ESD performance, please select suitable components for system protection.

LongCheer

Title	Memory_SD_SIM_Card
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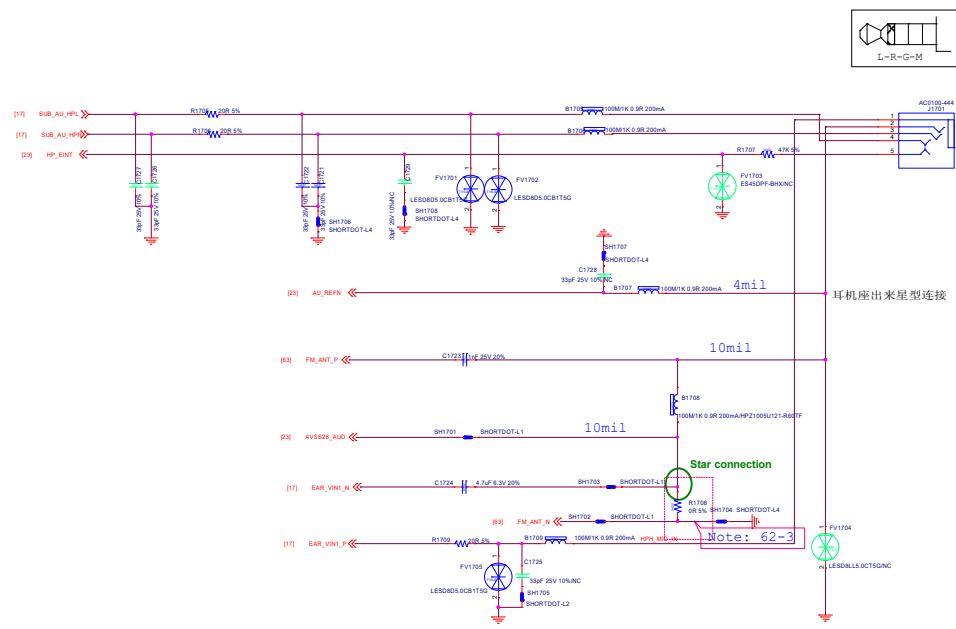
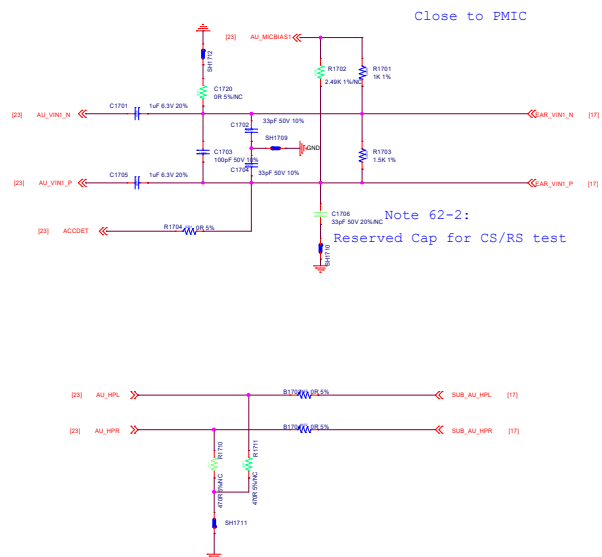
Size	Project	C3U_MB_V1
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Rev
V1

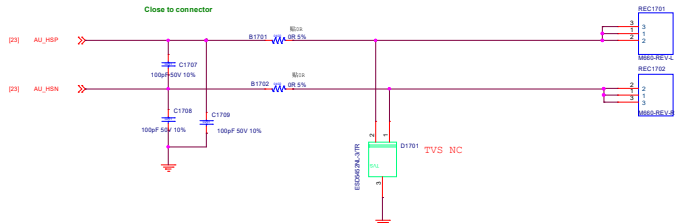
Date: Thursday, July 13, 2023

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Earphone Microphone



Receiver



Schematic design notice of "17_PERI_AUDIO_IO" page.

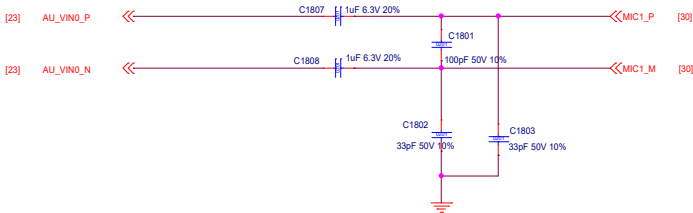
Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

Note 62-3:

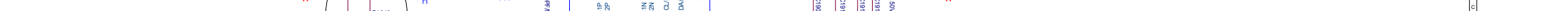
	Earphone Jack # Main board	Earphone Jack # Sub board
T1708	0 ohm	100nF

LongCheer			
Title PERI_AUDIO_IO			
Size D	Project C3U_MB_V1		Rev v1
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MIC -- Main



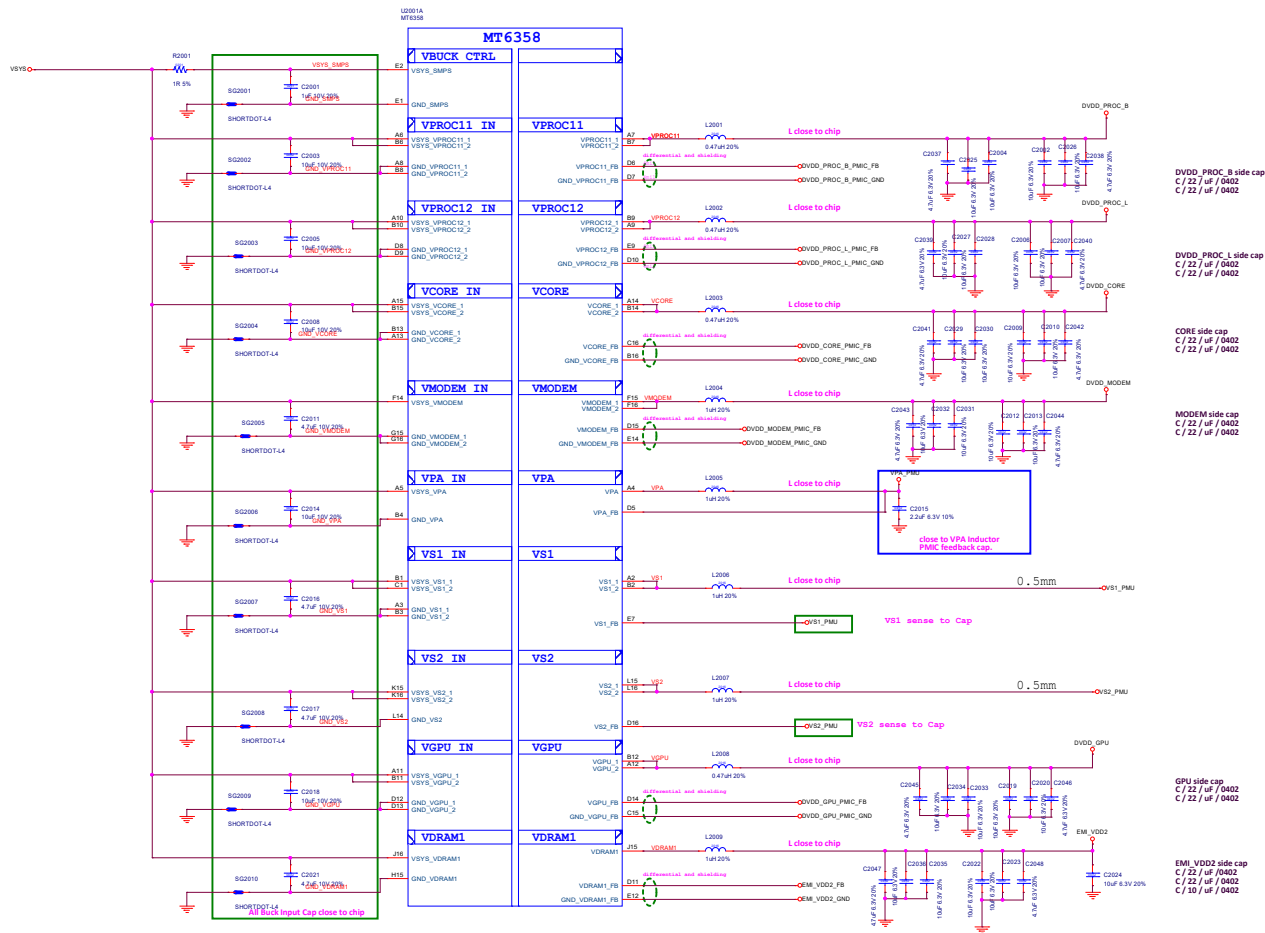
LongCheer			
Title 18_PERI_AUDIO_IO2(2 AMIC)			
Size C	Project C3U_MB_V1		Rev V1
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Title	PERI_AUDIO_IO2(2 AMIC)
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Project	C3U_MB_V1
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DVDD_PROC_B side cap
C/22 / uF / 0402
C/22 / uF / 0402

DVDD_PROC_L side cap
C/22 / uF / 0402
C/22 / uF / 0402

CORE side cap
C/22 / uF / 0402
C/22 / uF / 0402

MODEM side cap
C/22 / uF / 0402
C/22 / uF / 0402

GPU side cap
C/22 / uF / 0402
C/22 / uF / 0402

EMI_VDD2 side cap
C/22 / uF / 0402
C/22 / uF / 0402
C/10 / uF / 0402

LongCheer

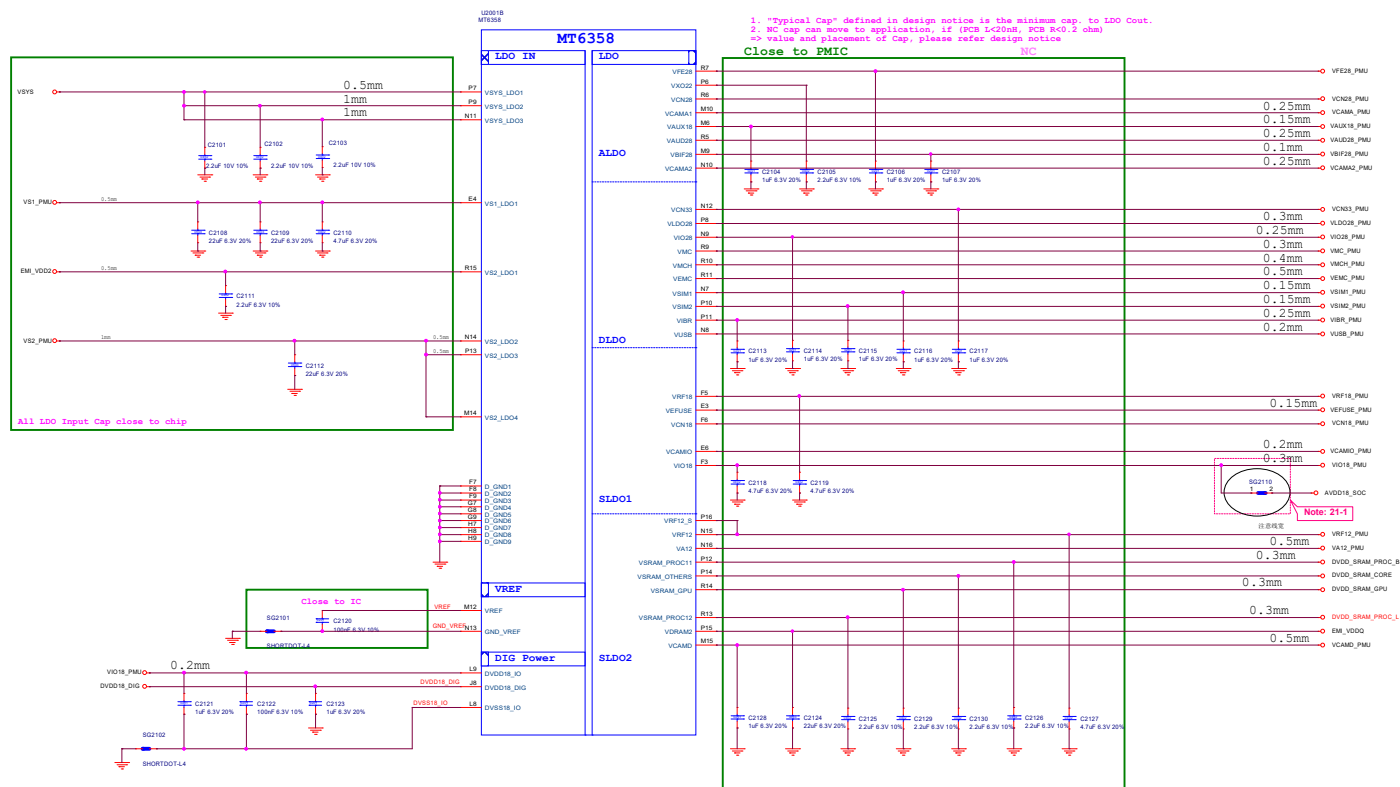
Title POWER_MT6358-Buck

Size D Project C3U_MB_V1

Rev v1

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Schematic design notice of "21_POWER_MT6358-LDO" page.

Note 21-1: Please set SG2110 close to C2118, making star connection between VIO18_PMU and AVDD18_SOC near to LDO cap. C2118
 Please also refer to MT6358 design notice for further detail design information

Note 21-2: If these power trace can meet LDO layout constraint, these CAP can be NC or removed.
 Please refer to MT6358 design notice.

LongCheer

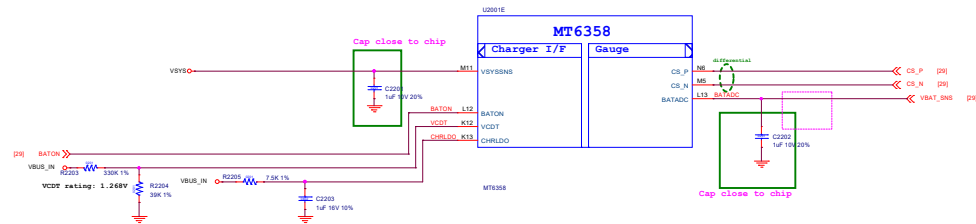
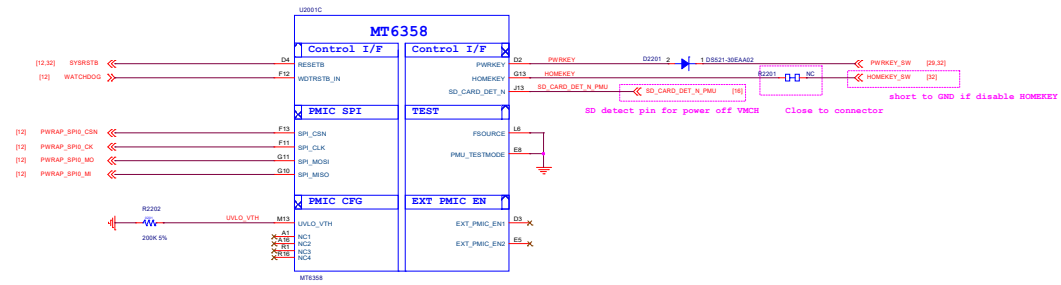
Title POWER_MT6358-LDO

Size D Project C3U_MB_V1

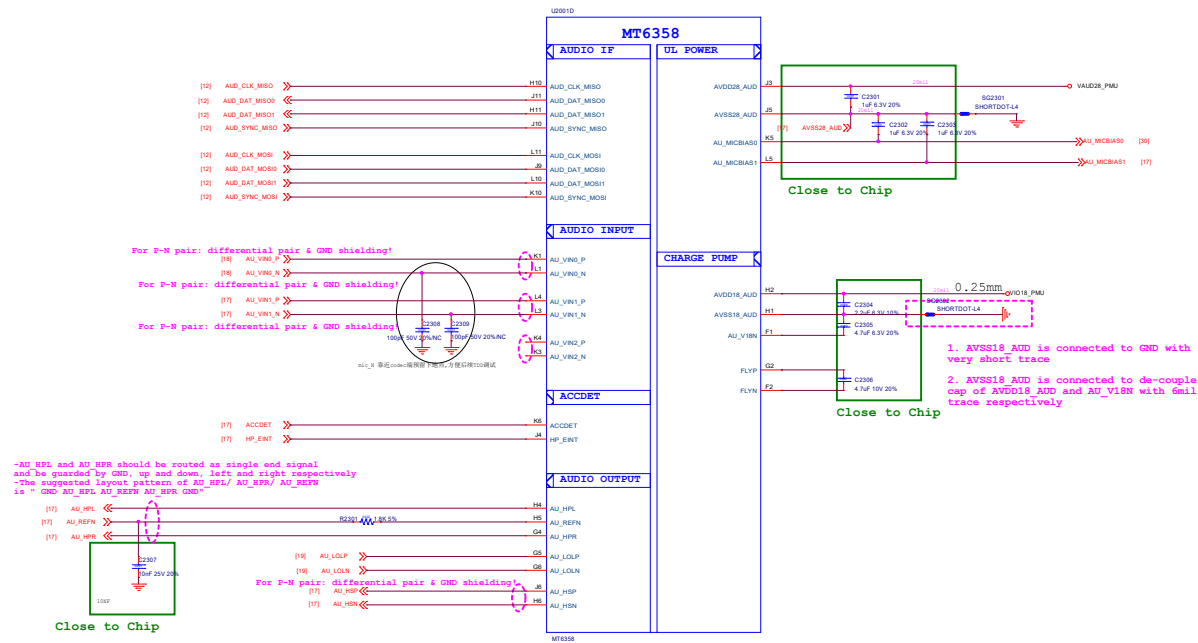
Rev V1

Date: Thursday, July 13, 2023

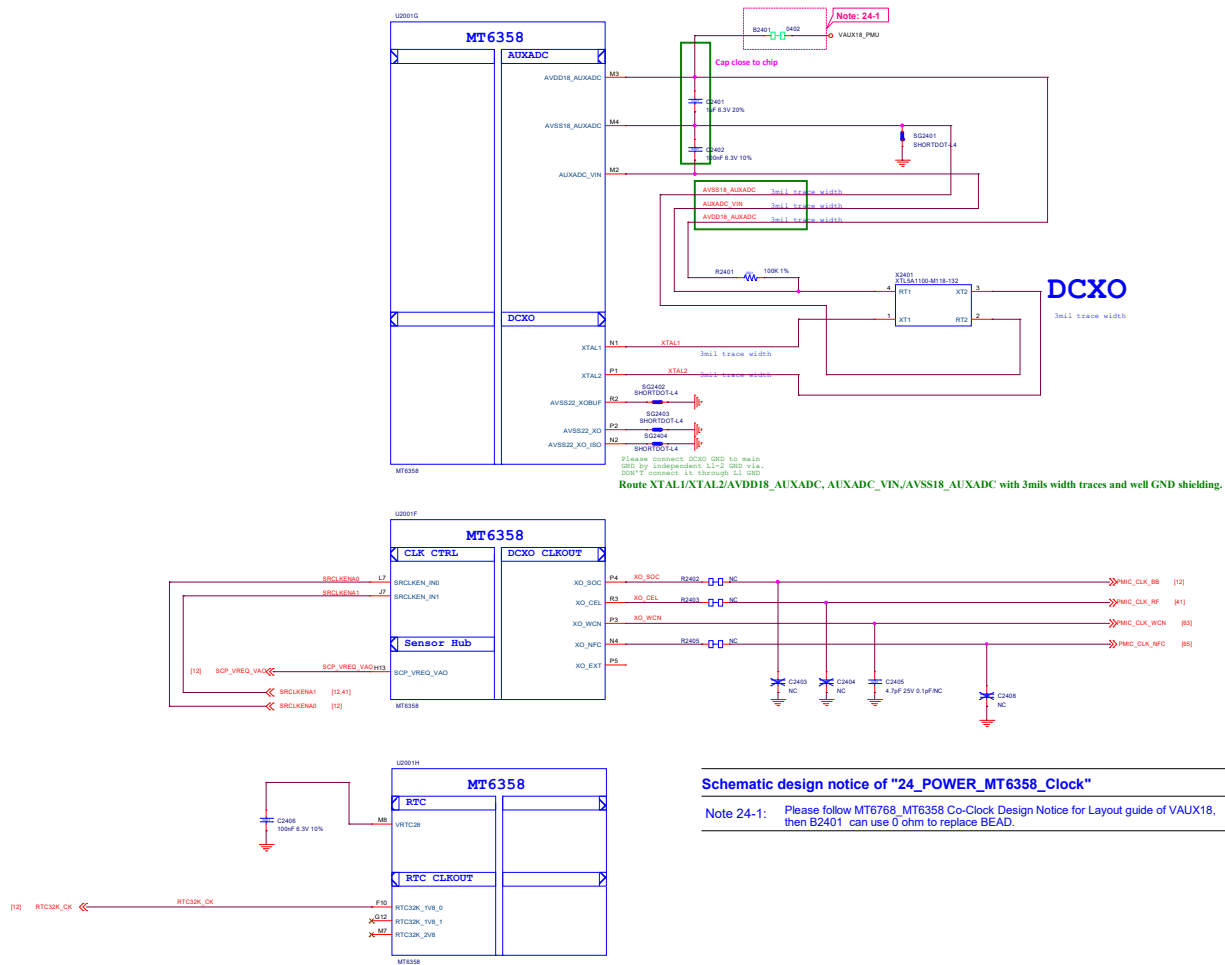
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LongCheer		
Title POWER_MT6358-IF		
Size D	Project C3U_MB_V1	Rev V1
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LongCheer		
Title POWER_MT6358-Audio		
Size D	Project C3U_MB_V1	Rev V1
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Schematic design notice of "24_POWER_MT6358_Clock"

Note 24-1: Please follow MT6768_MT6358 Co-Clock Design Notice for Layout guide of VAUX18, then B2401 can use 0 ohm to replace BEAD.

LongCheer

Title POWER_MT6358_Clock

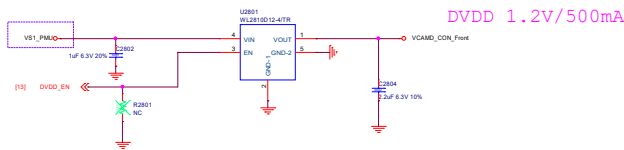
Size D Project C3U_MB_V1

Rev V1

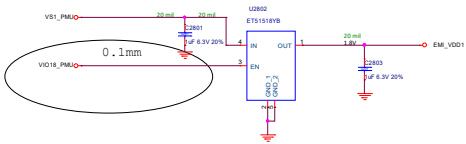
Date: Thursday, July 13, 2023

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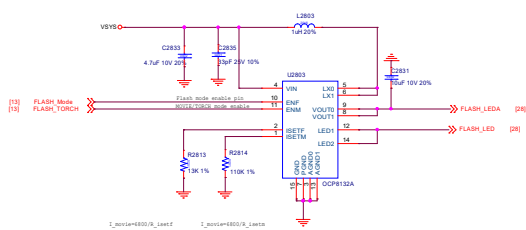
LDO for Front Camera (VCAMD_EXT1)
1.2V



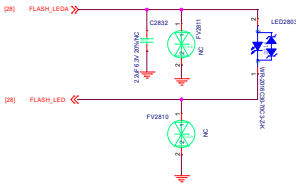
Ext. EMI_VDD1 for LPDDR4X VDD1



Flash LED Driver

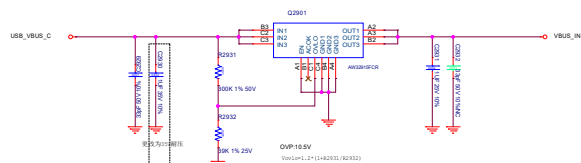


TVS默认不贴

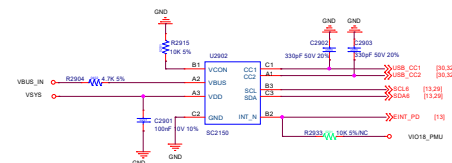


LongCheer		
Title POWERS_CAM_FL_DDR		
Size D	Project C3U_MB_V1	Rev v1
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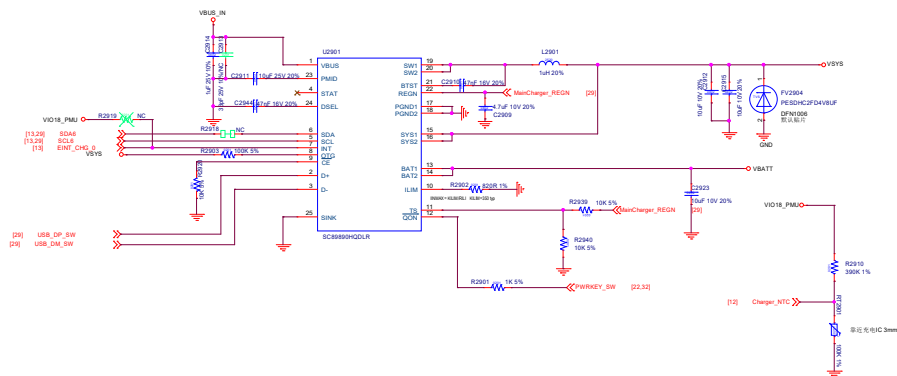
OVP



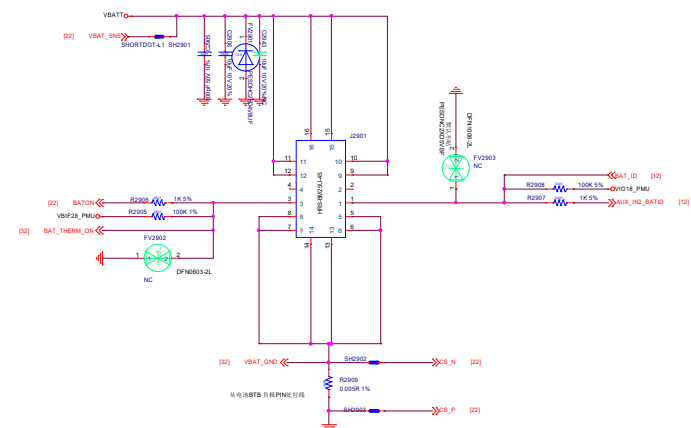
PD PHY



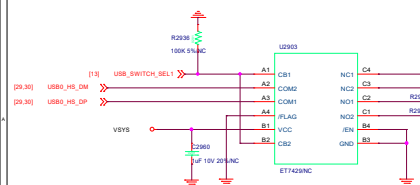
Switching Charger Power Path



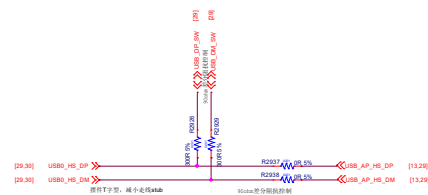
Battery Connector



USB Switch 预留

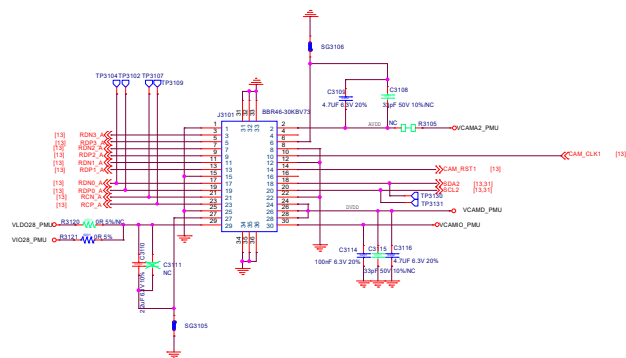


默认 DM DP T 型连接

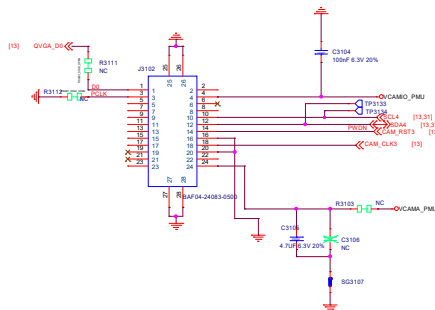


LongCheer		
Title EXT_BL+BIAS+CC+CHG+BAT		
Size D	Project C3U_MB_V1	Rev v1
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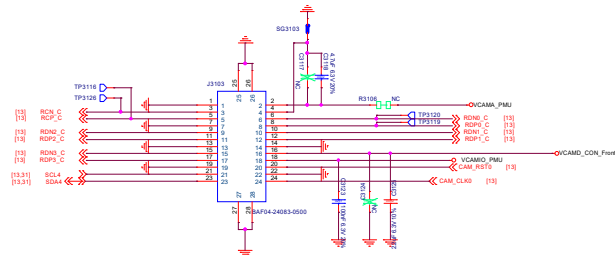
Main Camera 50M



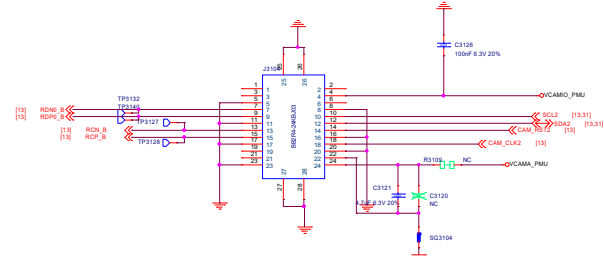
QVGA Camera



FRONT CAMERA



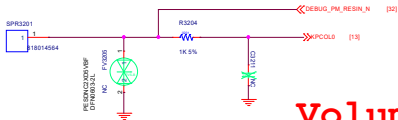
Marco Camera 2M



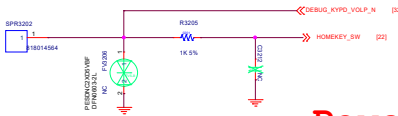
LongCheer			
Title PERI_CAMERA			
Size D	Project C3U_MB_V1		Rev v1
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KEY BTB

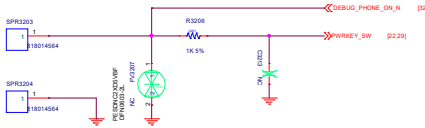
Volume Down



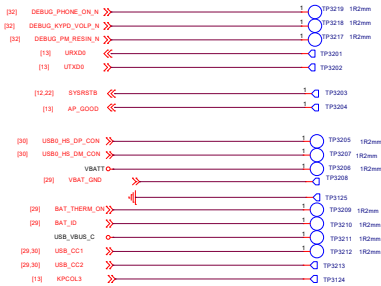
Volume Up



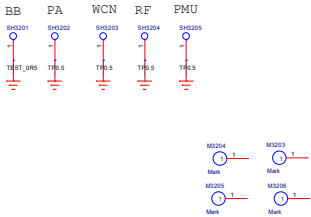
Power On



Test Point

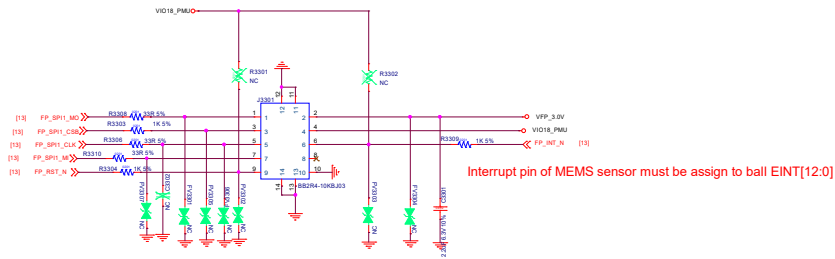


Shielding

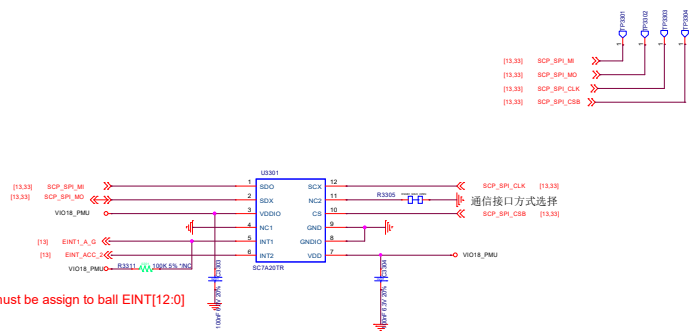


LongCheer		
Title Conn_key_Debug_io		
Size D	Project C3U_MB_V1	Rev V1
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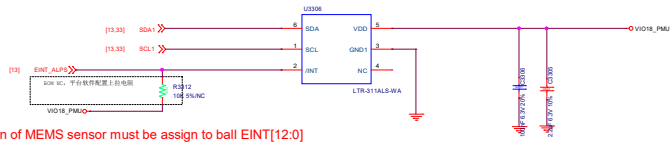
FP-Sensor



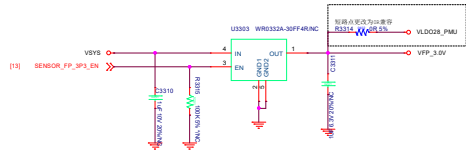
A-Sensor



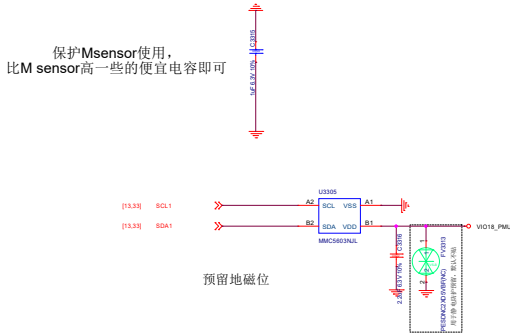
L-Sensor



指纹LDO预留(NC)



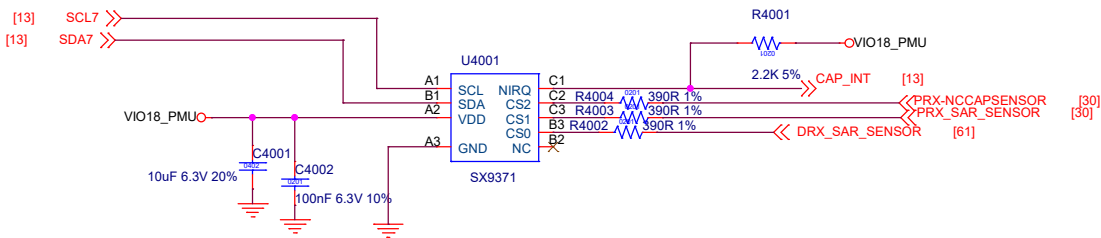
Compass



[M sensor] Keep a minimum distance of 15mm from power ICs / PCB traces of more than 100mA / magnet component. Check HW design notice for more detail

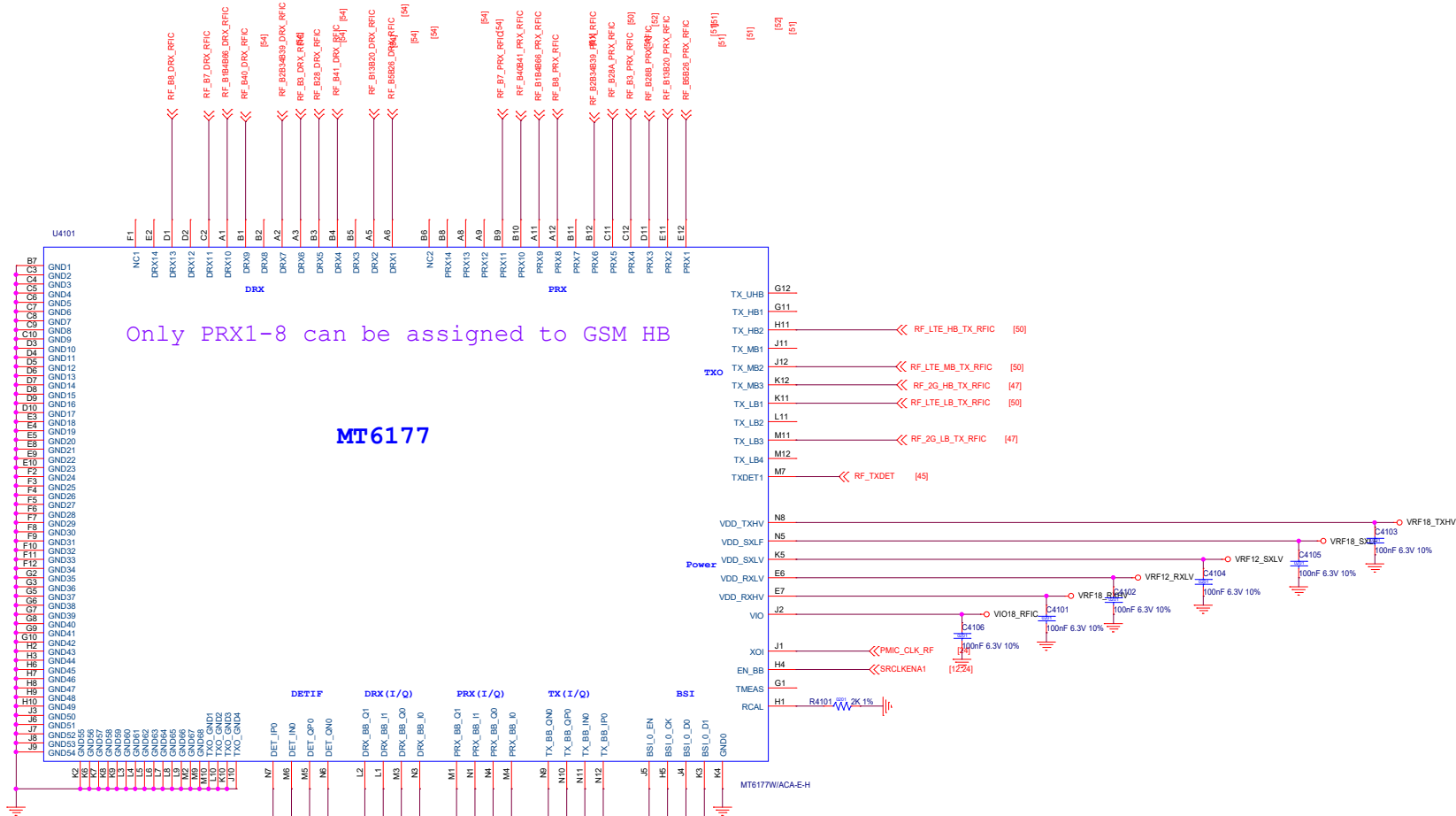
LongCheer		
Title PERI_SENSORS_MEMs_ALS/PS		
Size D	Project C3U_MB_V1	Rev v1
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SAR sensor

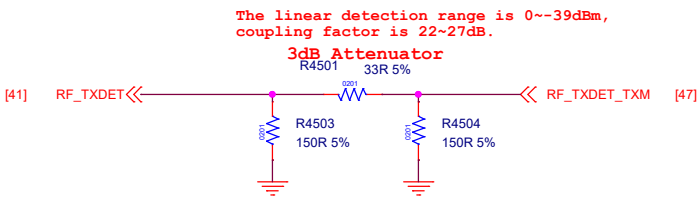


Title		
40_RF_SARSensor		
Size	Document Number	Rev
B	<Doc>	1.0
Date:	Thursday, July 13, 2023	Sheet 40 of 66

MT6177_WMSS

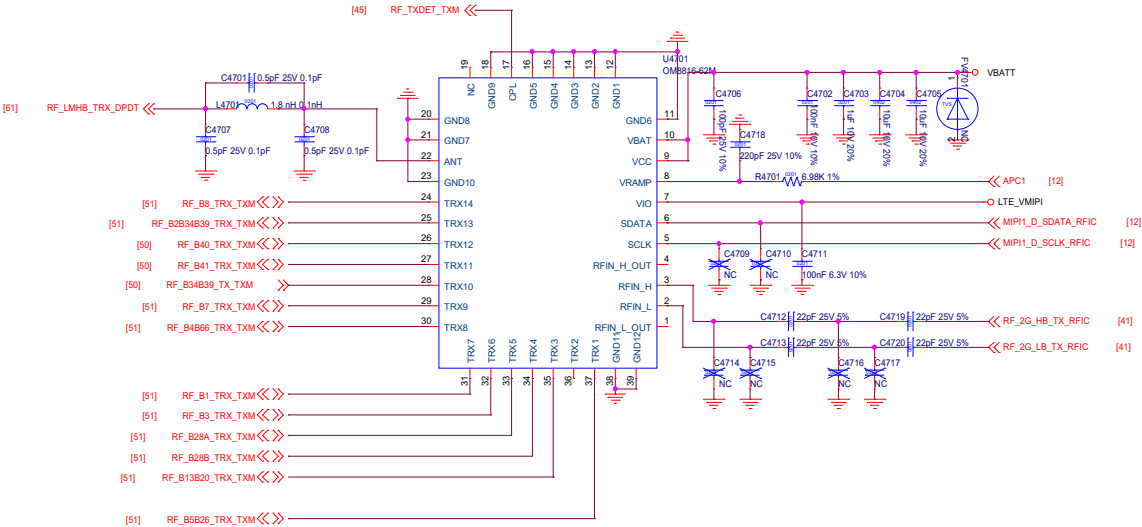


RF_FBRX

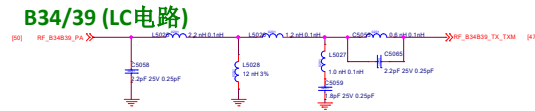
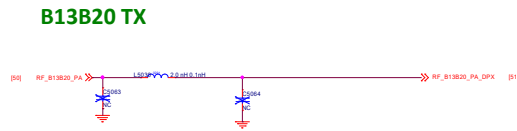
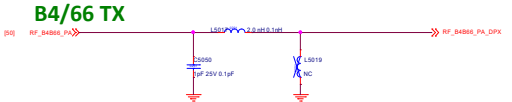
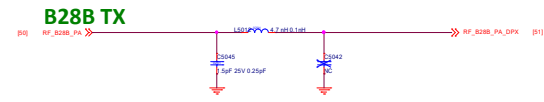
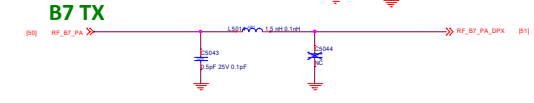
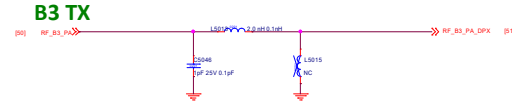
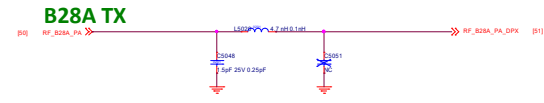
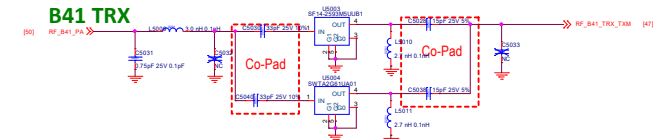
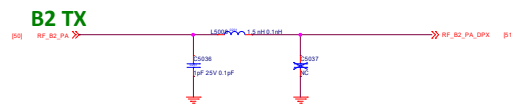
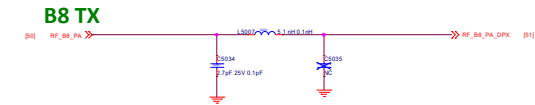
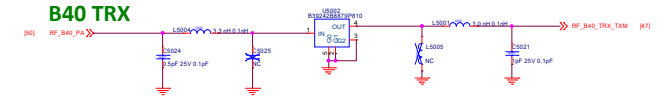
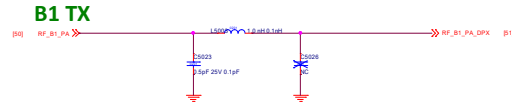
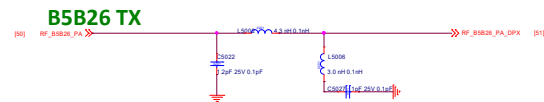
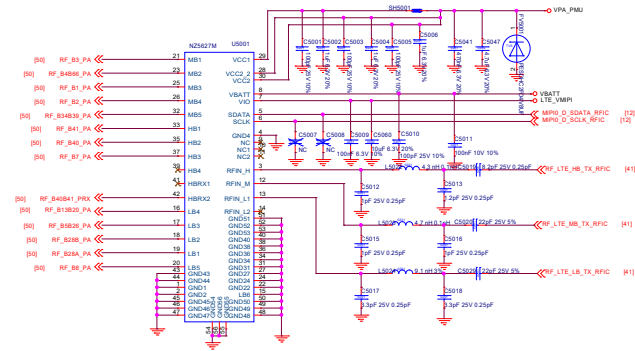


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RF_TXM

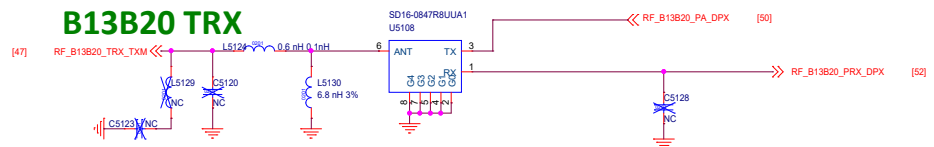
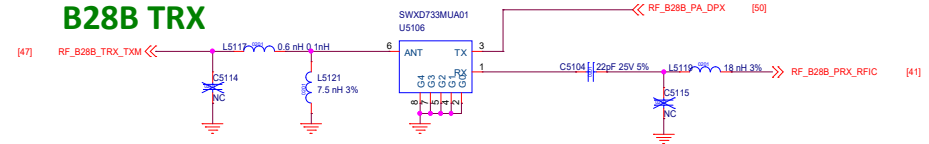
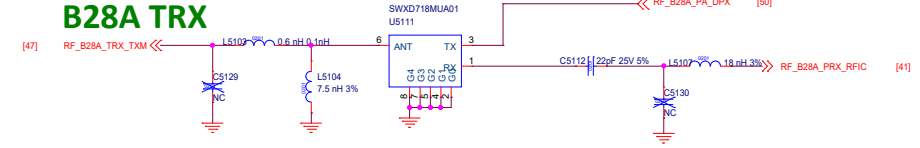
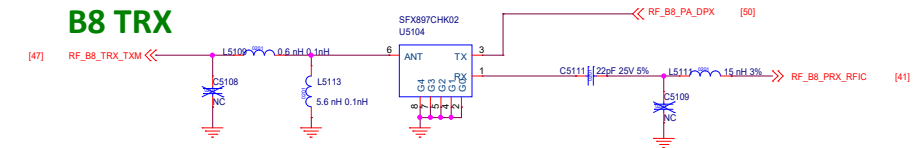
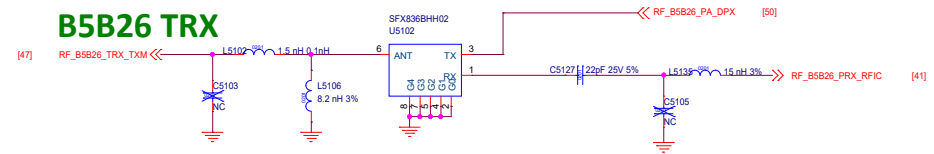
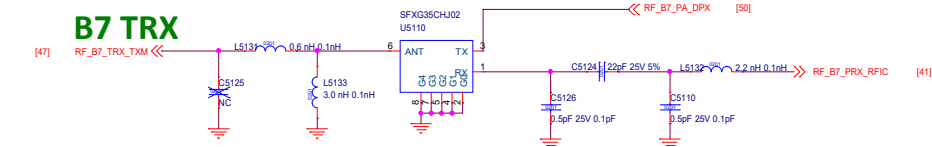
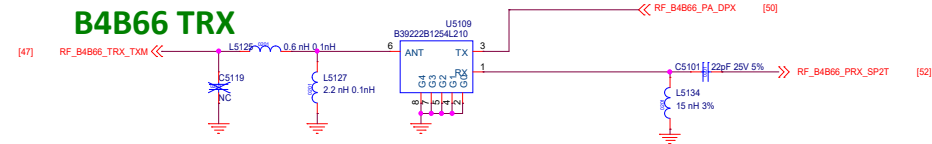
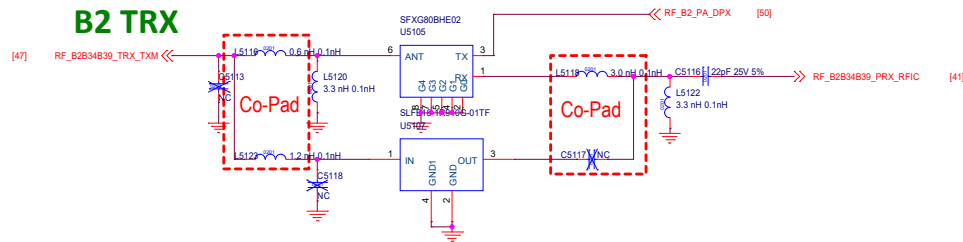
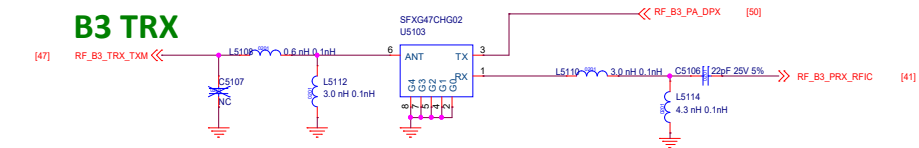
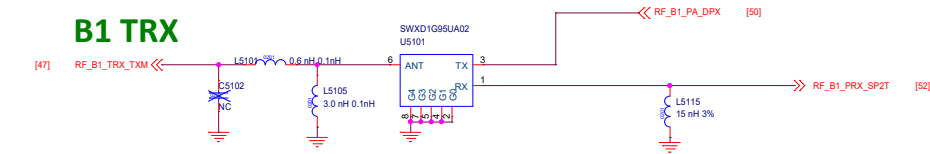


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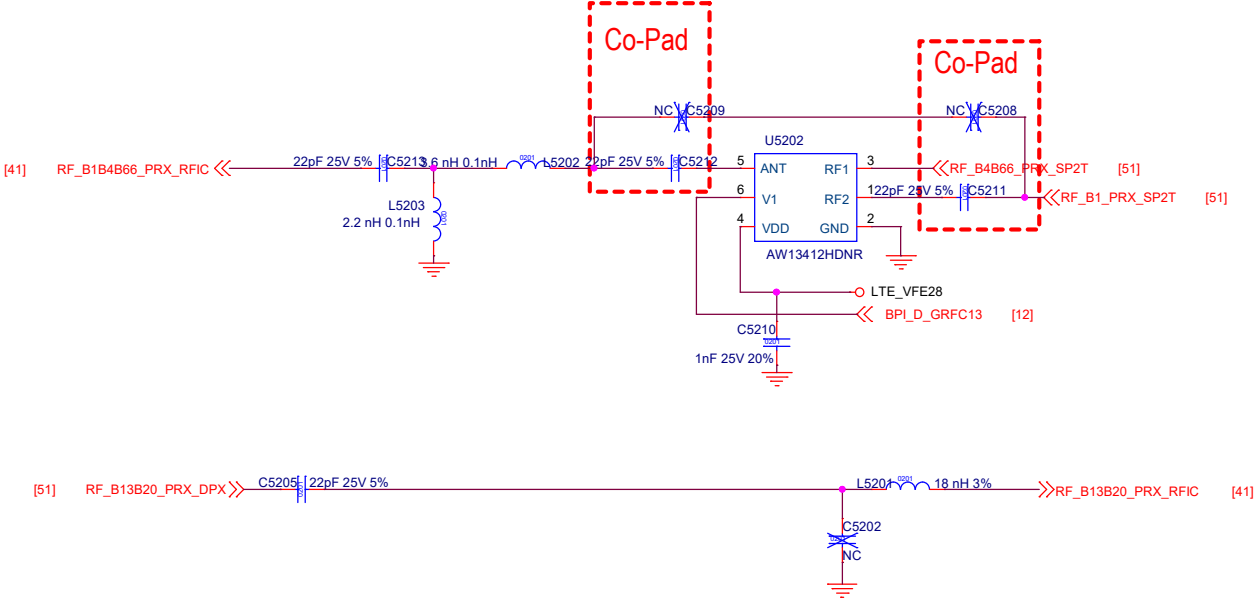


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50_RF_ENDC_PA		
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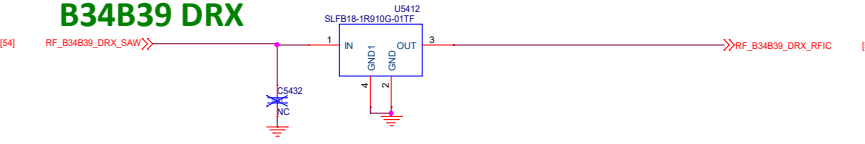
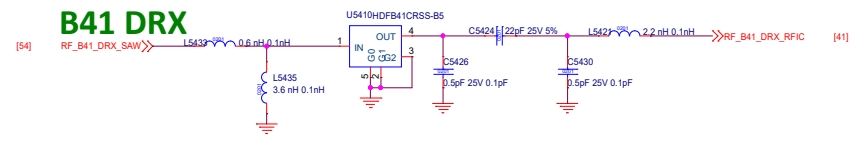
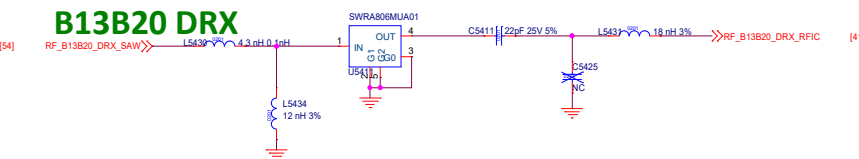
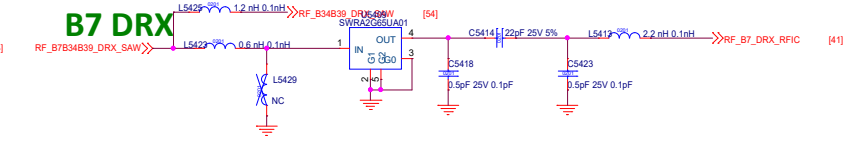
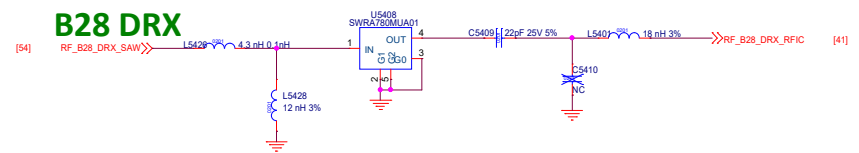
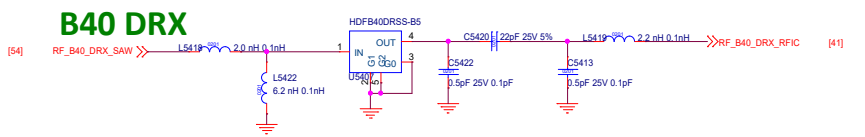
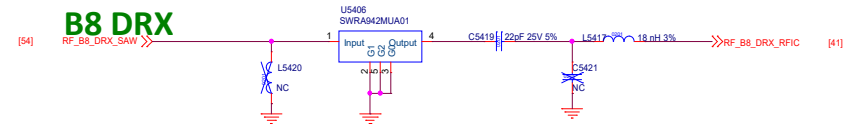
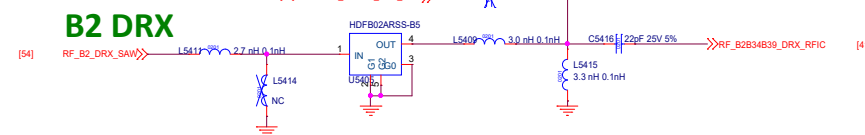
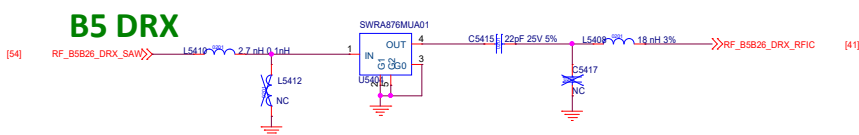
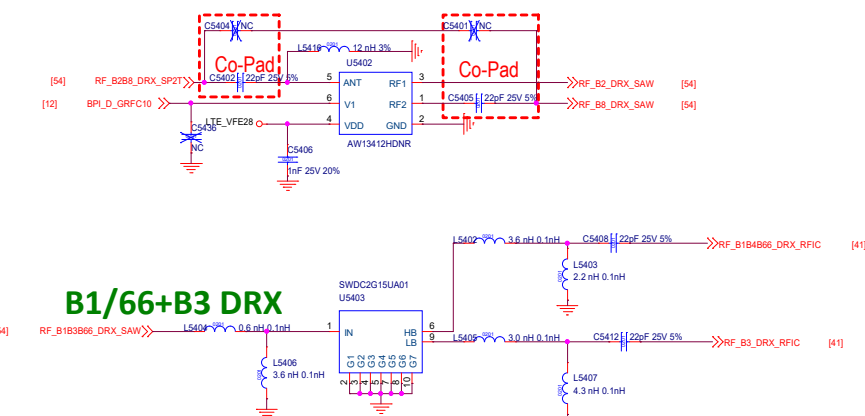
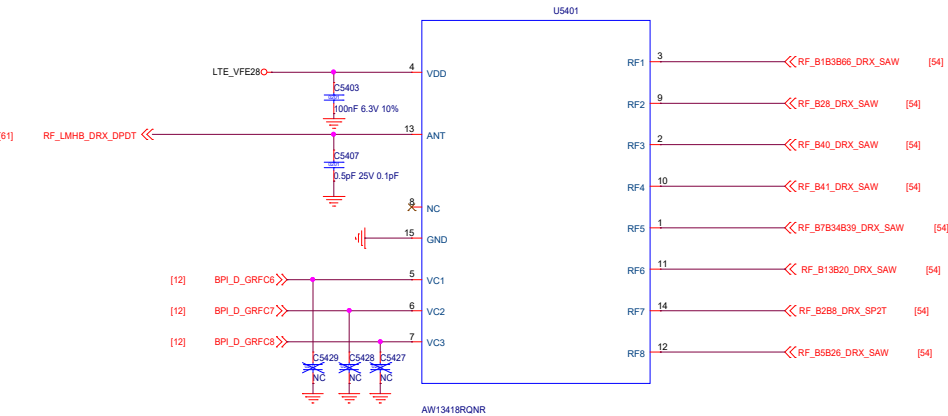


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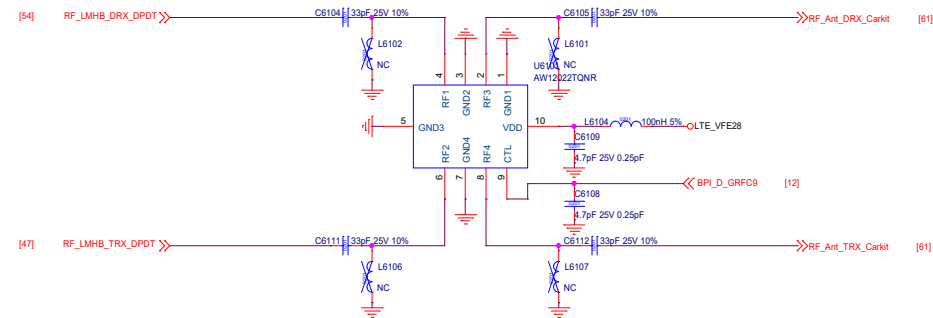
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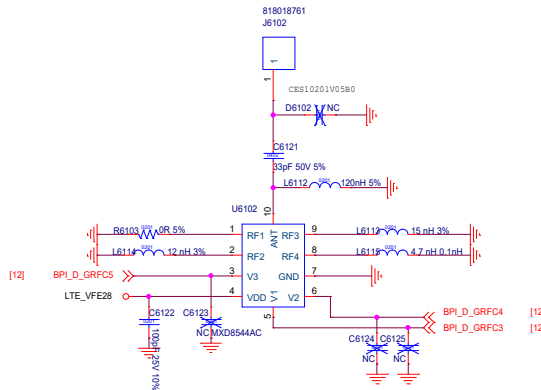
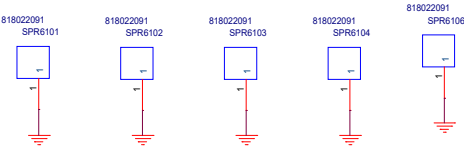
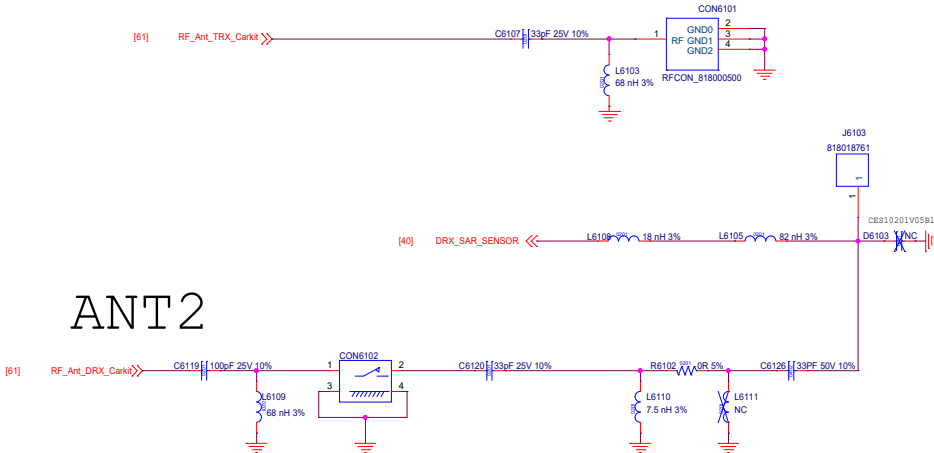
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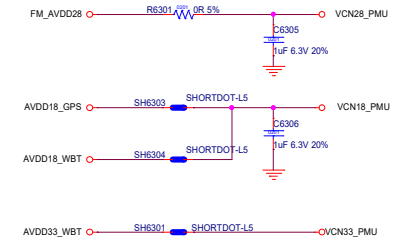
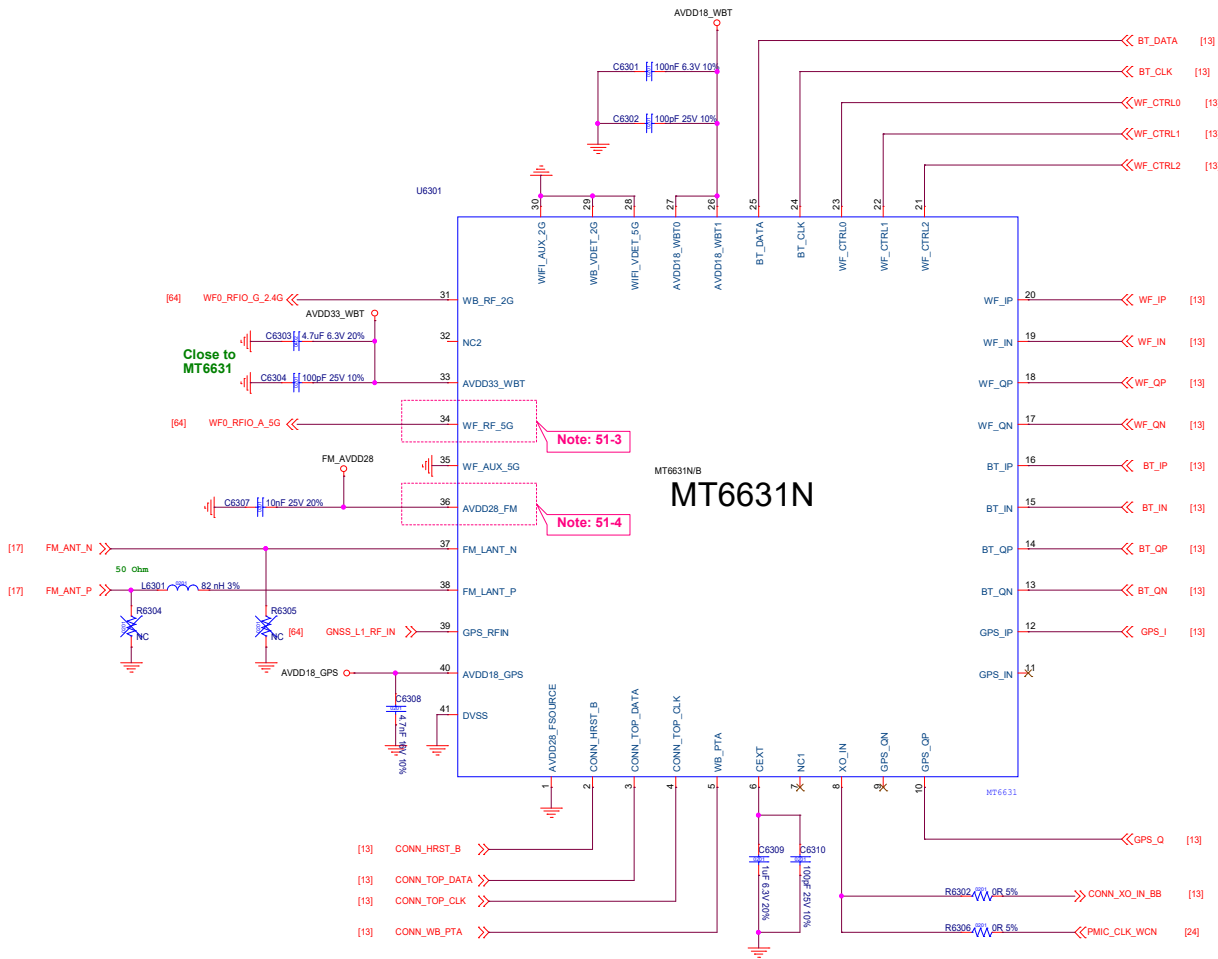
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ANT2



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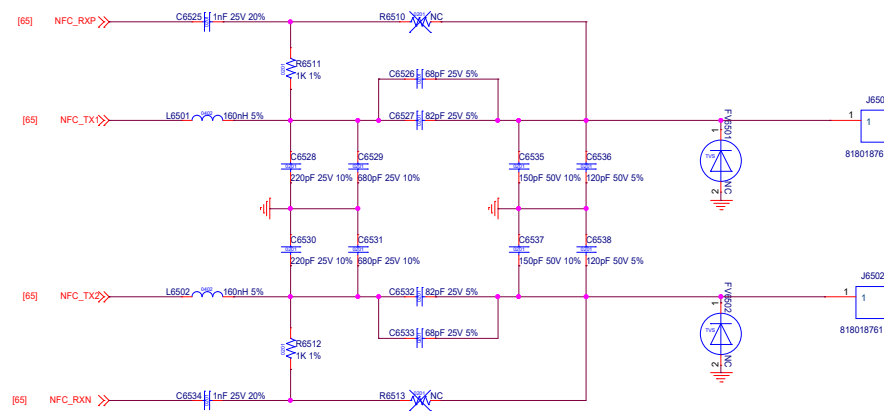
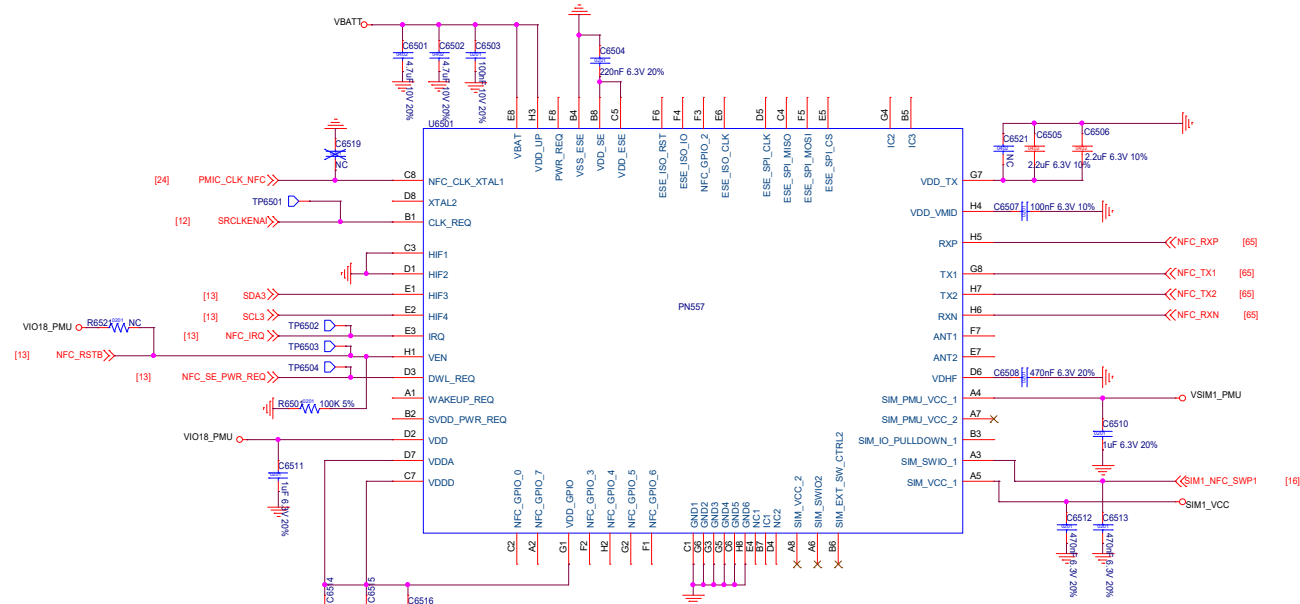
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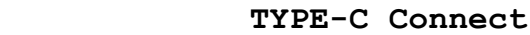
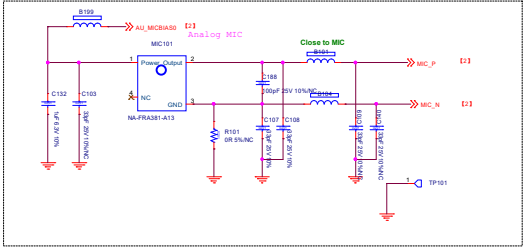
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P0.2 0414: 02页， 马达增加D201器件

P0.2 0424: 02页， MIC增加C288 B299

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